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<!DOCTYPE html>
<html lang="de">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>FKT Dyson-Simulation V4.3</title>
  <script src="https://cdn.tailwindcss.com"></script>
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <style>
    body { background-color: #0f172a; color: #e2e8f0; font-family: 'Inter', sans-serif; }
    .card { background: rgba(30, 41, 59, 0.7); backdrop-filter: blur(10px); border: 1px solid
    rgba(255,255,255,0.1); border-radius: 1rem; }
    .glow { box-shadow: 0 0 20px rgba(56, 189, 248, 0.2); }
  </style>
</head>
<body class="p-4 md:p-8">
  <div class="max-w-6xl mx-auto">
    <header class="mb-8 text-center">
      <h1 class="text-4xl font-bold text-sky-400 mb-2">FKT
      Dyson-Effizienz-Simulator</h1>
      <p class="text-slate-400">Numerische Simulation der Bulk-Ableitung pro Gramm
      Flerovium (V4.3)</p>
    </header>

    <div class="grid grid-cols-1 lg:grid-cols-3 gap-8">
      <!-- Controls -->
      <div class="card p-6 glow">
        <h2 class="text-xl font-semibold mb-4 text-sky-300 border-b border-sky-900
        pb-2">Parameter</h2>

        <div class="space-y-6">
          <div>
            <label class="block text-sm mb-2">GMA-Effizienz-Faktor (Standard:
            120x)</label>
            <input type="range" id="gmaFactor" min="1" max="250" value="120"
            class="w-full h-2 bg-slate-700 rounded-lg appearance-none cursor-pointer accent-sky-500">
            <div class="flex justify-between text-xs mt-1"><span>1x</span><span>
            id="gmaVal" class="text-sky-400 font-bold">120x</span><span>250x</span></div>
          </div>

          <div>
            <label class="block text-sm mb-2">Flerovium-Masse (Gramm)</label>
            <input type="number" id="flMass" value="1.0" step="0.1" class="w-full
            bg-slate-800 border border-slate-600 rounded p-2 text-sky-300">
          </div>

          <div>
            <label class="block text-sm mb-2">Solare Einstrahlung (Relativ)</label>

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        <input type="range" id="solarInput" min="1" max="100" value="50"
class="w-full h-2 bg-slate-700 rounded-lg appearance-none cursor-pointer
accent-amber-500">
        <div class="flex justify-between text-xs mt-1"><span>Min</span><span
id="solarVal">50%</span><span>Max (L<sub>#9737</sub>)</span></div>
        </div>

        <div class="p-3 bg-sky-900/30 rounded text-xs text-sky-200 border
border-sky-700">
        <strong>Info:</strong> Die Simulation nutzt die Resonanzfrequenz von 3,773
MeV und den nuklearen Anker  $\eta$ <sub>Dim</sub>.
        </div>
    </div>
    </div>
    <!-- Visualization -->
    <div class="lg:col-span-2 space-y-8">
        <div class="card p-6 glow h-full">
            <h2 class="text-xl font-semibold mb-4 text-sky-300">Energie-Ableitung:
Thermisch vs. Bulk</h2>
            <canvas id="efficiencyChart" height="200"></canvas>

            <div class="grid grid-cols-2 md:grid-cols-4 gap-4 mt-8">
                <div class="text-center">
                    <span class="block text-xs text-slate-400
uppercase">Bulk-Ableitung</span>
                    <span id="bulkOut" class="text-2xl font-bold text-sky-400">0.00
MW</span>
                </div>
                <div class="text-center">
                    <span class="block text-xs text-slate-400 uppercase">Therm. Last</span>
                    <span id="thermOut" class="text-2xl font-bold text-rose-400">0.00
MW</span>
                </div>
                <div class="text-center">
                    <span class="block text-xs text-slate-400 uppercase">Verschränkung
(<math>\Phi</math>)</span>
                    <span id="phiDensity" class="text-2xl font-bold text-emerald-400">0.00
<math>\rho</math></span>
                </div>
                <div class="text-center">
                    <span class="block text-xs text-slate-400 uppercase">Status</span>
                    <span id="statusLabel" class="text-lg font-bold">Stabil</span>
                </div>
            </div>
        </div>
    </div>
    </div>
    </div>
    </div>

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<!-- System Architecture Logic -->
<div class="mt-8 card p-6 border-sky-500/30">
  <h3 class="text-lg font-bold text-sky-300 mb-4 text-center italic">Analyse der Typ-II
Skalierung</h3>
  <p class="text-sm leading-relaxed text-slate-300">
    Bei einem GMA-Faktor von <span class="gmaPlaceholder font-bold">120</span>
reduziert sich die thermische Signatur der Kollektoren drastisch. Das Flerovium agiert als
"Phasen-Konverter", der die solaren Photonen über die 3,773 MeV Resonanz in den
Bulk-Raum leitet. Dadurch können Dyson-Kollektoren deutlich kompakter gebaut werden, da
die Kühlung nicht mehr der limitierende Faktor ist. Das Ziel ist eine <strong>Geometrische
Kohärenz von > 98%</strong> für den verlustfreien Quanten-Transfer.
  </p>
</div>
</div>

<script>
const ctx = document.getElementById('efficiencyChart').getContext('2d');
let chart;

const inputs = {
  gma: document.getElementById('gmaFactor'),
  mass: document.getElementById('filMass'),
  solar: document.getElementById('solarInput')
};

const outputs = {
  bulk: document.getElementById('bulkOut'),
  therm: document.getElementById('thermOut'),
  phi: document.getElementById('phiDensity'),
  status: document.getElementById('statusLabel'),
  gmaLabels: document.querySelectorAll('.gmaPlaceholder')
};

function calculate() {
  const gma = parseFloat(inputs.gma.value);
  const mass = parseFloat(inputs.mass.value);
  const solar = parseFloat(inputs.solar.value) / 100;

  // FKT Konstanten
  const etaDim = 2.54; // Log-Skala Anker
  const resonanceEnergy = 3.773; // MeV

  // Basisleistung (Beispielwert für Simulation normiert)
  const basePower = solar * 500 * mass;

  // Die Bulk-Ableitung steigt mit GMA-Faktor und Resonanzkopplung
  const bulkPower = basePower * (gma / 100) * (resonanceEnergy / 2);

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const thermPower = basePower / (gma / 10);

const phi = (bulkPower / (mass * etaDim)).toFixed(3);

// UI Update
outputs.bulk.innerText = bulkPower.toFixed(2) + " MW";
outputs.therm.innerText = thermPower.toFixed(2) + " MW";
outputs.phi.innerText = phi + " ρ";
document.getElementById('gmaVal').innerText = gma + "x";
document.getElementById('solarVal').innerText = (solar * 100).toFixed(0) + "%";
outputs.gmaLabels.forEach(el => el.innerText = gma);

if (thermPower > 50) {
  outputs.status.innerText = "Kritische Hitze";
  outputs.status.className = "text-lg font-bold text-rose-500";
} else {
  outputs.status.innerText = "Kausal Stabil";
  outputs.status.className = "text-lg font-bold text-emerald-500";
}

updateChart(bulkPower, thermPower);
}

function updateChart(bulk, therm) {
  const data = {
    labels: ['Bulk-Ableitung (MW)', 'Thermische Last (MW)'],
    datasets: [{
      data: [bulk, therm],
      backgroundColor: ['#38bdf8', '#fb7185'],
      borderWidth: 0,
      borderRadius: 10
    }]
  };

  if (chart) {
    chart.data = data;
    chart.update();
  } else {
    chart = new Chart(ctx, {
      type: 'bar',
      data: data,
      options: {
        indexAxis: 'y',
        responsive: true,
        plugins: { legend: { display: false } },
        scales: {
          x: { grid: { color: 'rgba(255,255,255,0.05)' }, ticks: { color: '#94a3b8' } },
          y: { grid: { display: false }, ticks: { color: '#f8fafc', font: { weight: 'bold' } } }
        }
      }
    });
  }
}

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    }  
  }  
});  
}  
  
[inputs.gma, inputs.mass, inputs.solar].forEach(input => {  
  input.addEventListener('input', calculate);  
});  
  
window.onload = calculate;  
</script>  
</body>  
</html>
```